

COSC 4370 Homework #3

Overview:

Continuing with our work on OpenGL, we would like to be able to load an object from a file and then add some lighting to the object. In this assignment, we will also be working with a classic Graphics object, the Utah Teapot. We read the vertices and faces from a file and construct the teapot using triangles. This file contains a list of all the vertices in the file followed by all the faces. We will be using a special library to read the object file and provide the faces and vertices we need. We will then add lighting to the teapot. To do so, we will need to know the surface vectors for the triangular faces. We will compute these normal vectors using the vertices of each face.

Specifications:

Start with the python code example given that shows how to add lighting to a cube. Add in code to read in the vertices and faces of the teapot from the **teapot.obj** file. We would like to display a simply shaded teapot as the final result. Read in the vertices and faces of the teapot, compute the normal vector for each face, construct the teapot, and then light the teapot. It should be upright and tilted toward the camera and rotate about its vertical axis.

Deliverables:

- Submit a single python file showing the lighted teapot. You will not need to include the obj file for the teapot itself. Do not modify the .obj file.
- The program should display a flat-shaded teapot with a blue light up and left of the camera and a red light down and right of the camera.
- You may use any color for the teapot itself. It should be upright on the screen and rotate to show off the spout and handle.
- Use a camera perspective or orthographic projection that includes the whole teapot rotating in the scene to show off the shading.

Notes:

- You will need to compute the normal to each of the faces in the teapot. You can do so by taking the cross-product of the first two edges of each triangle. You can work out the computations by hand or use the numpy library to help you.
- Be sure to start this program early as there are many small details.